

Where MarinFold goes wrong on protein complexes

A contact-map atlas | 17 FoldBench test complexes | Six diagnostic case studies

Usually wrong contacts; sometimes no interface prediction.

20.3%

emit no inter-chain contacts

61.6%

emit scorable contacts, all wrong

16.7%

include at least one correct pair

Other attempts: 1.4% predict only unscorable interface pairs; 8 / 17,000 are unfinished.

“At least one correct” can still mean a largely wrong map. Categories refer to whole rollouts.

What to look for

1. Experimental vs frequency heatmaps: do frequent predictions land on the right interface patch?
2. Top-R error maps: which high-vote pairs are wrong, and which true pairs are missed?
3. A typical rollout vs oracle best@1000: do correct contacts ever occur together?

Six cases are purposefully chosen to span observed behaviors; they are not a random sample.

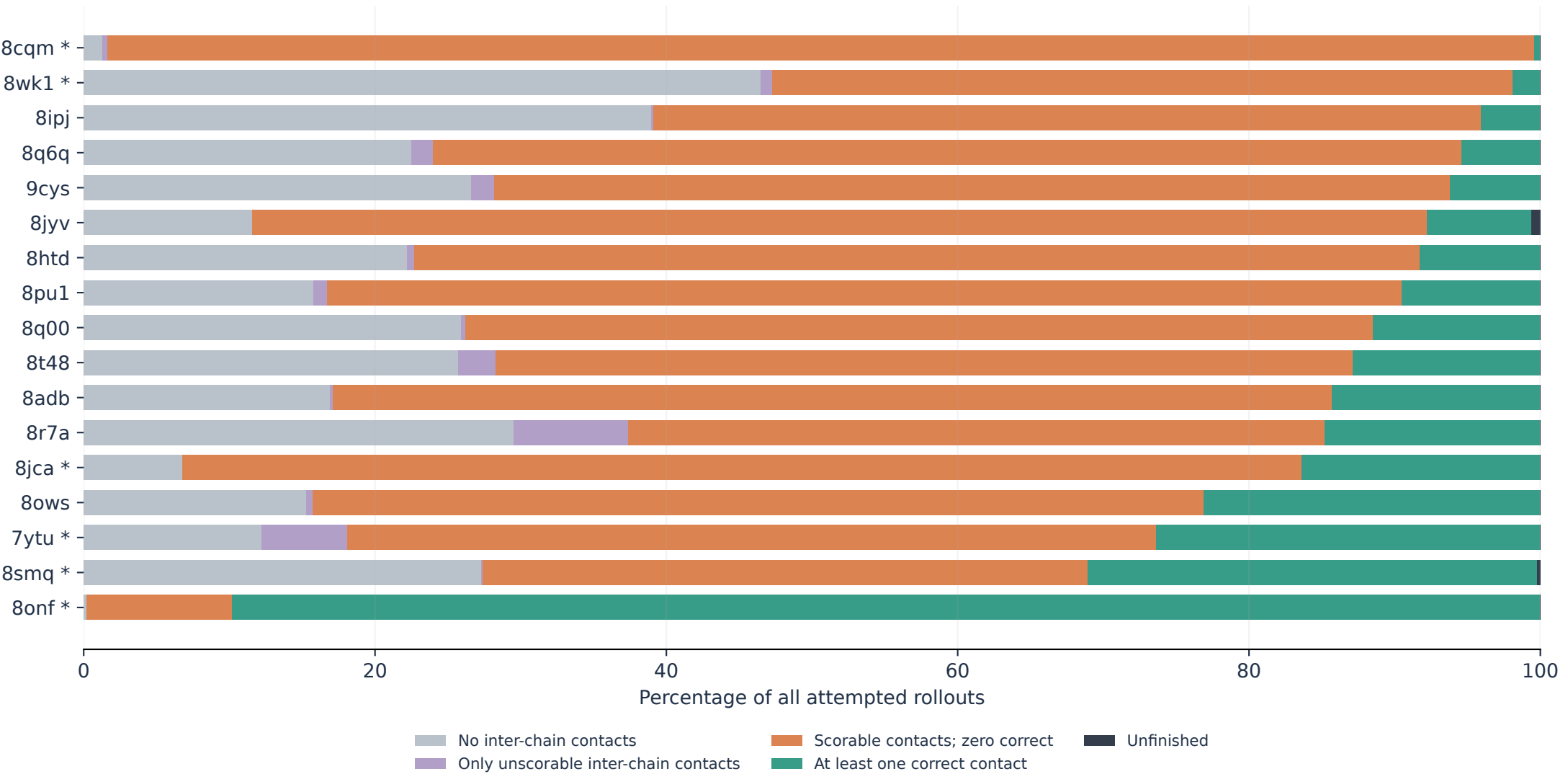
Cohort summaries always include all 17 test complexes. No additional inference was run.

The best@1000 map uses experimental truth for selection. It is a diagnostic ceiling.

How often is a rollout empty, wrong, or partly right?

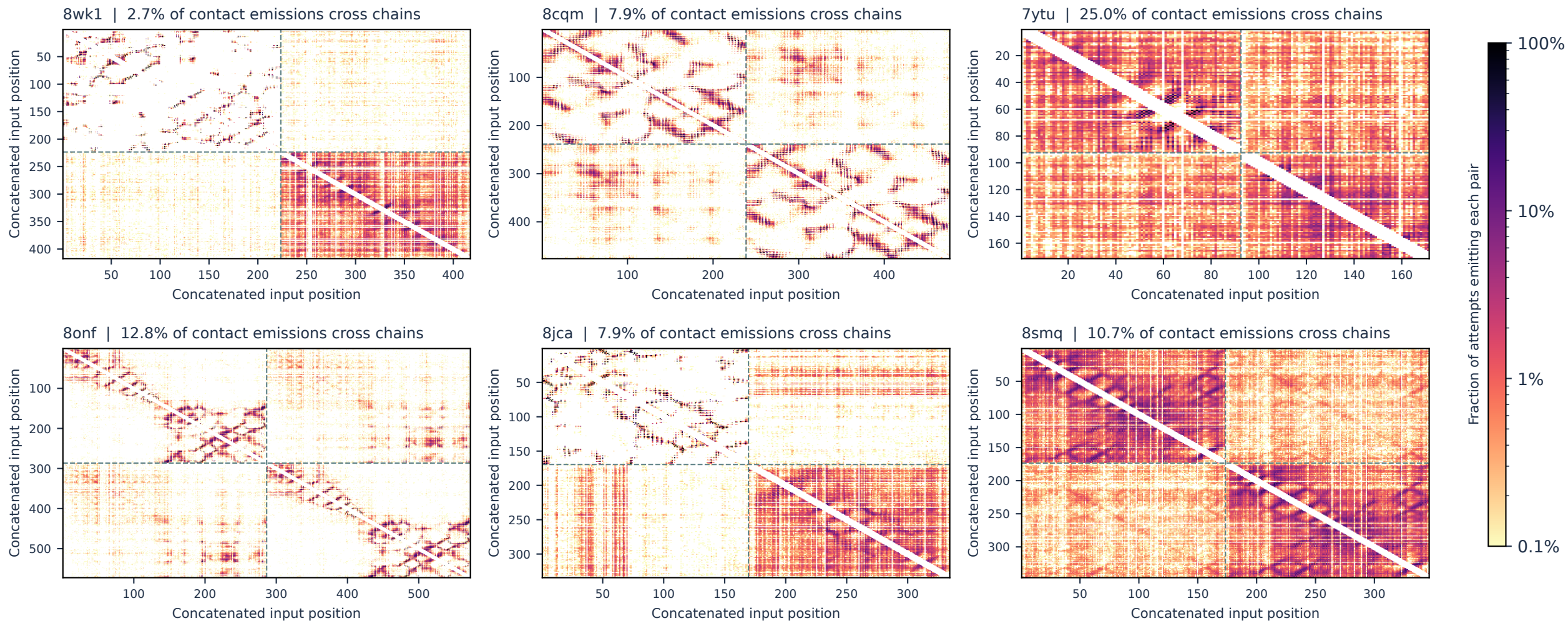
Each bar represents 1,000 attempts. A partly right map contains at least one true contact; it can contain many false contacts.

* Included in the detailed case studies



The model emits many more intra-chain than inter-chain contacts

Full two-chain prediction matrices; dashed boundaries separate chains. These panels show output allocation, not intra-chain accuracy.



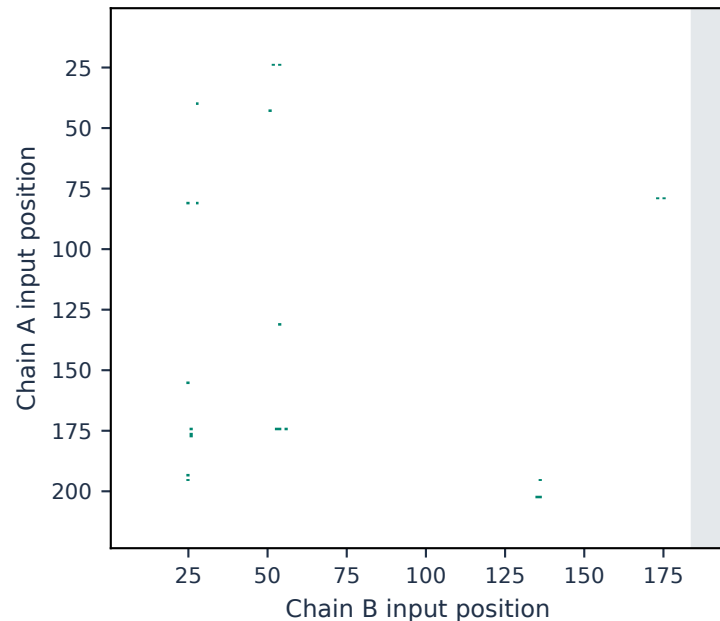
All emitted contacts are shown here, including positions unresolved in the experiment. The next pages score resolved cross-chain pairs only.

8wk1 | Often emits no interface contacts

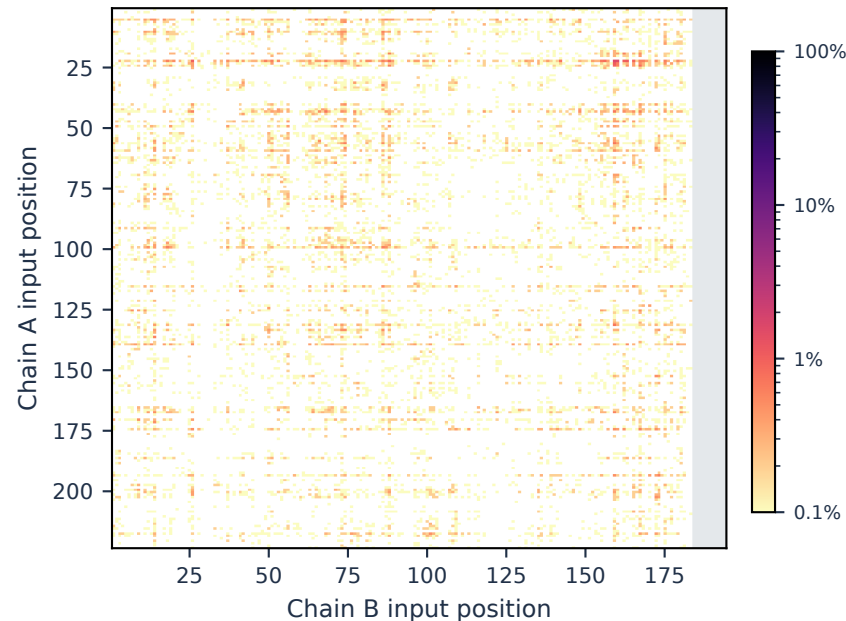
Bovine trypsin in complex with Durio zibethinus trypsin inhibitor DzTI-4

heterodimer | Chains 223 + 194 residues | 21 experimental interface contacts | Test split

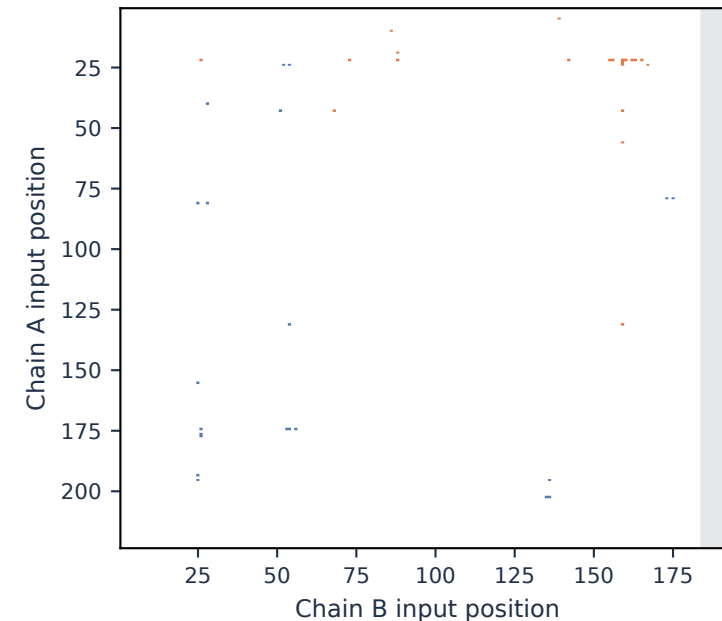
Experimental interface
21 true residue pairs



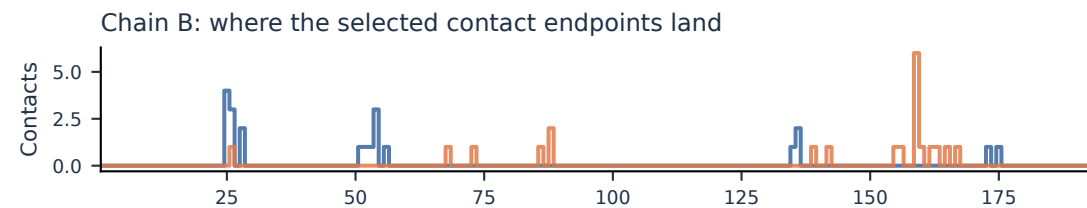
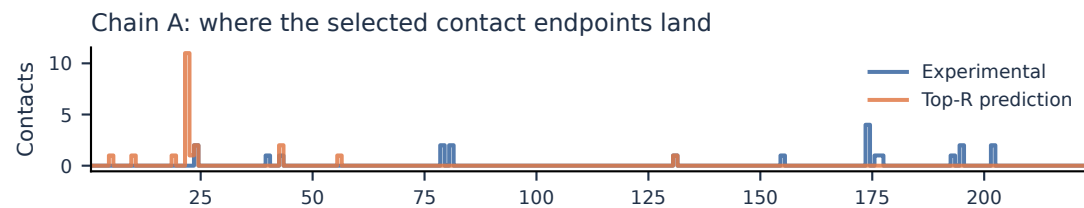
MarinFold contact frequency
All 1,000 attempted rollouts



Highest-vote 21 pairs (top-R)
0 correct, 21 wrong, 21 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable

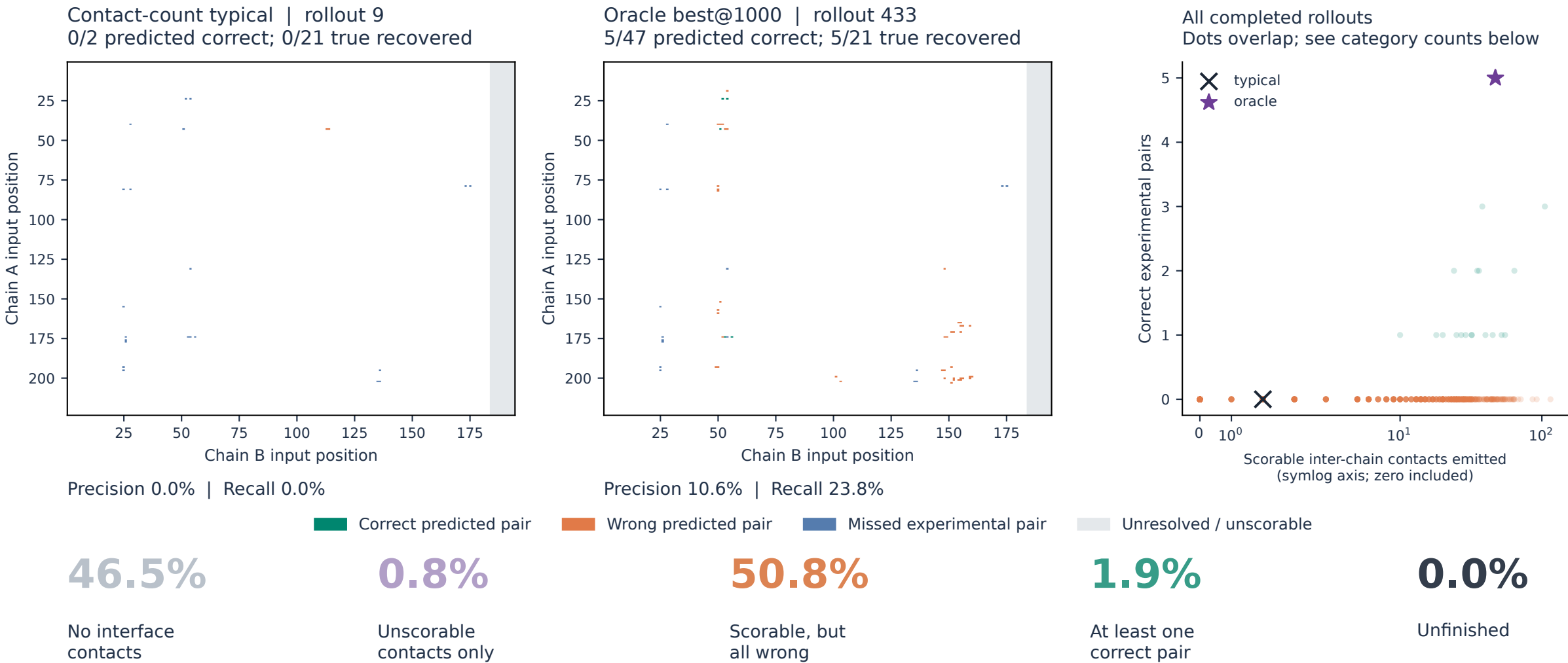


Frequent silence combines with poorly localized contacts when an interface is emitted.

Top-R precision = recall = 0.0%. Predicted endpoint residues on the true interface: A 33%, B 7%. True pairs ever sampled: 14/21.

8wk1 | What happens inside individual rollouts?

Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.



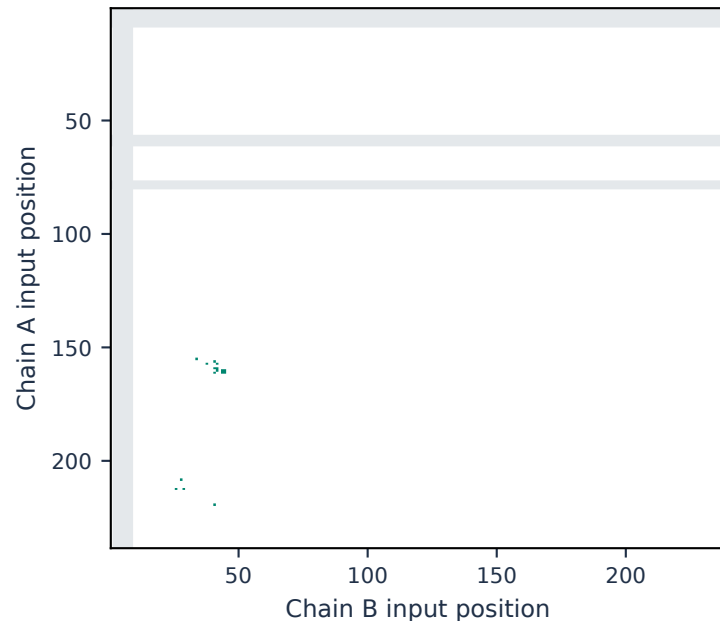
All-rollout median scorable contact count (completed only): 2. Experimental contact count: 21. A green pair is correct; a green-containing rollout is not necessarily a good map.

8cqm | Almost always predicts the wrong interface

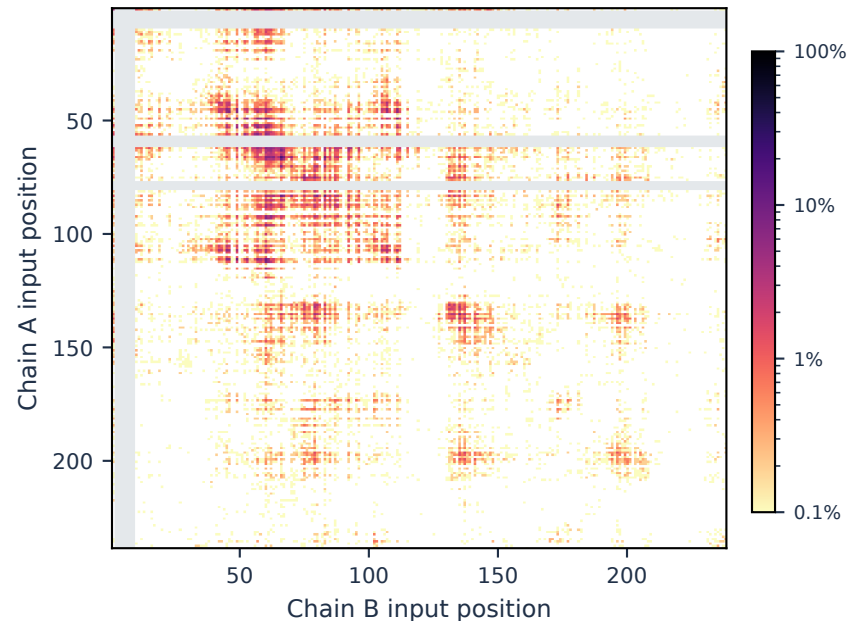
Broad-range phospholipase C from *Listeria monocytogenes*

homodimer | Chains 238 + 238 residues | 16 experimental interface contacts | Test split

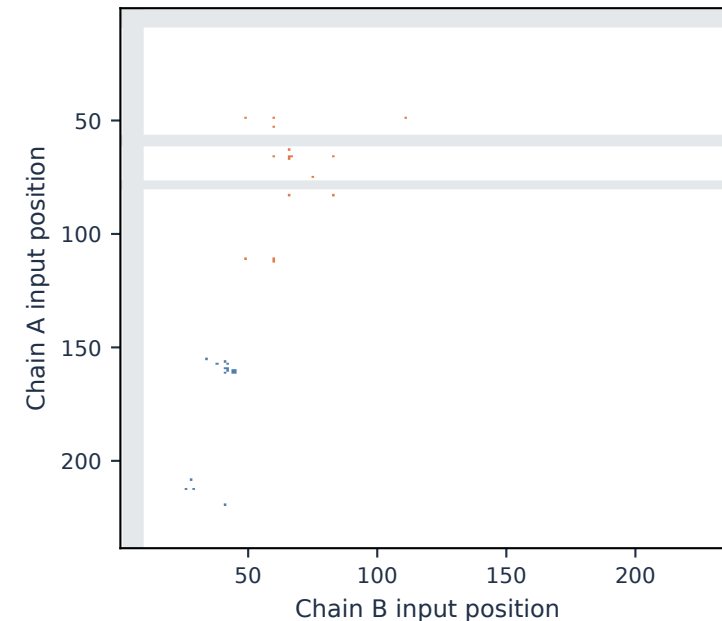
Experimental interface
16 true residue pairs



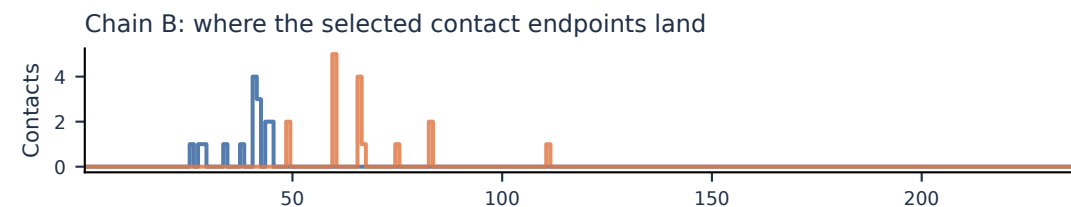
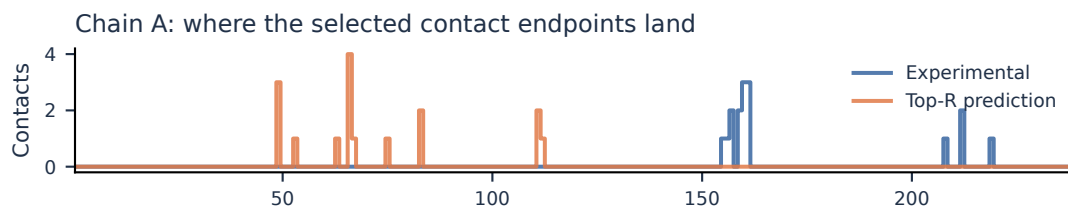
MarinFold contact frequency
All 1,000 attempted rollouts



Highest-vote 16 pairs (top-R)
0 correct, 16 wrong, 16 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable



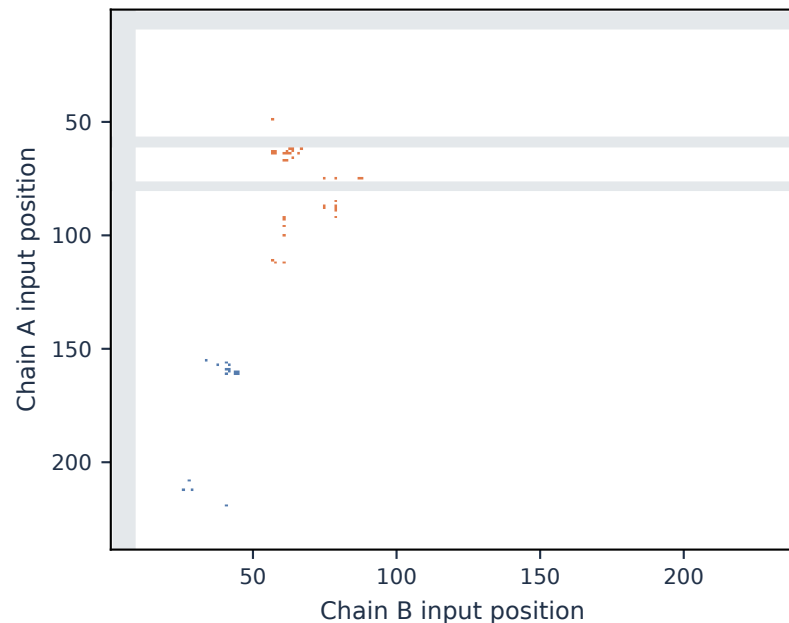
Top-ranked contact endpoints miss the experimental interface on both chains.

Top-R precision = recall = 0.0%. Predicted endpoint residues on the true interface: A 0%, B 0%. True pairs ever sampled: 5/16.

8cqm | What happens inside individual rollouts?

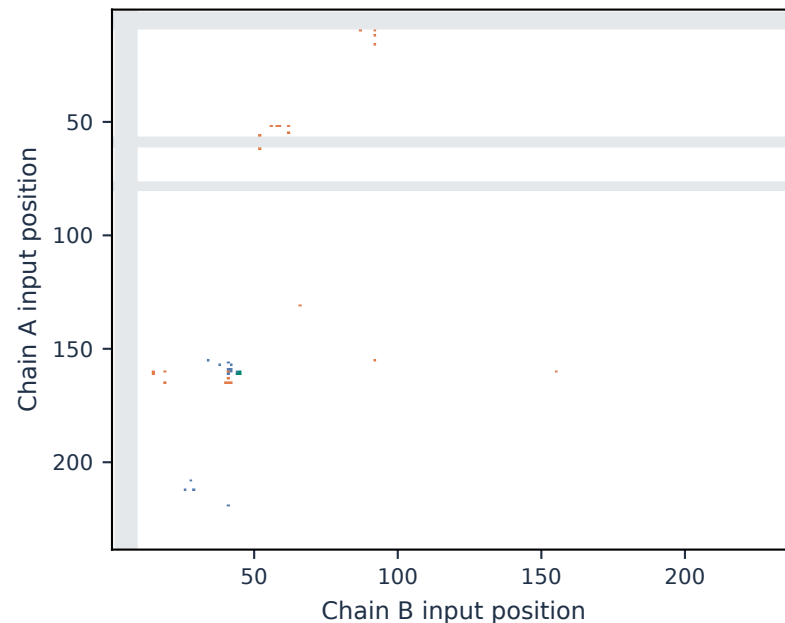
Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.

Contact-count typical | rollout 48
0/35 predicted correct; 0/16 true recovered



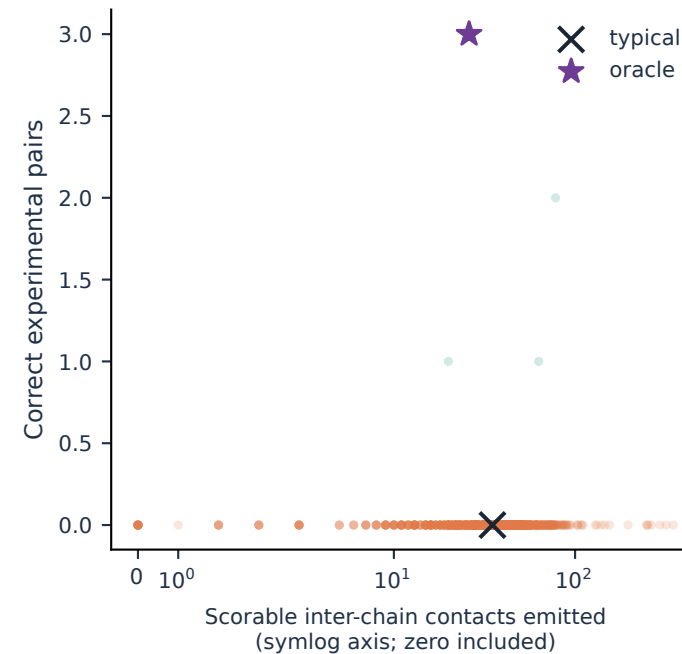
Precision 0.0% | Recall 0.0%

Oracle best@1000 | rollout 44
3/26 predicted correct; 3/16 true recovered



Precision 11.5% | Recall 18.8%

All completed rollouts
Dots overlap; see category counts below



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable

1.3%

0.3%

98.0%

0.4%

0.0%

No interface
contacts

Unscorable
contacts only

Scorable, but
all wrong

At least one
correct pair

Unfinished

All-rollout median scorable contact count (completed only): 35. Experimental contact count: 16. A green pair is correct; a green-containing rollout is not necessarily a good map.

7ytu | Correct contacts appear, but lose the vote

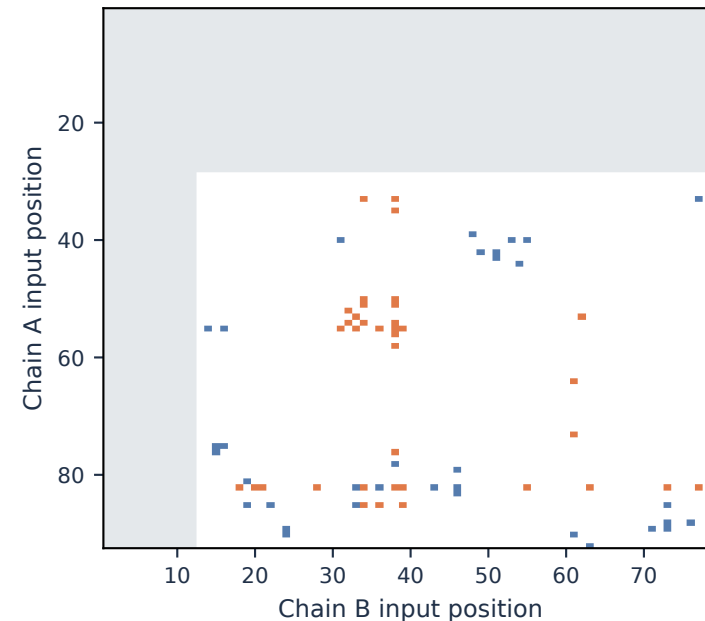
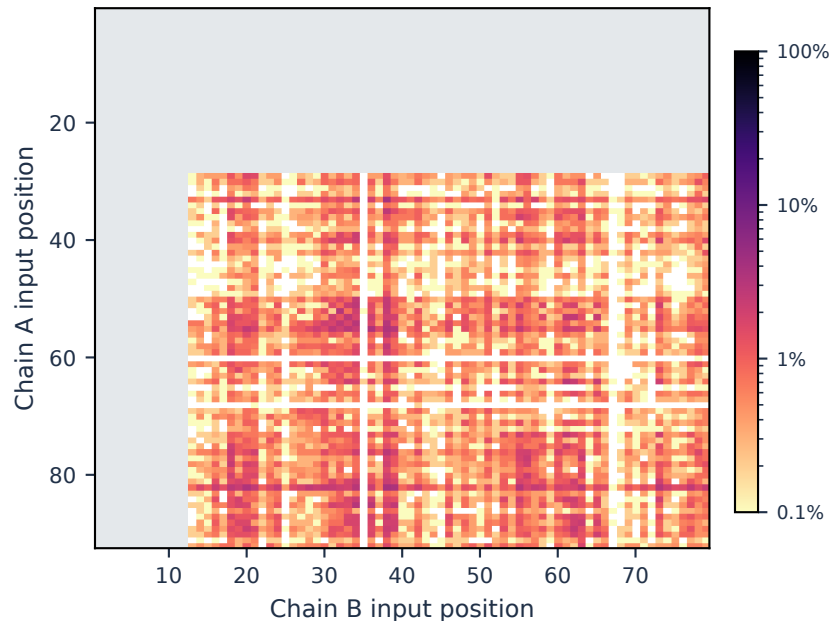
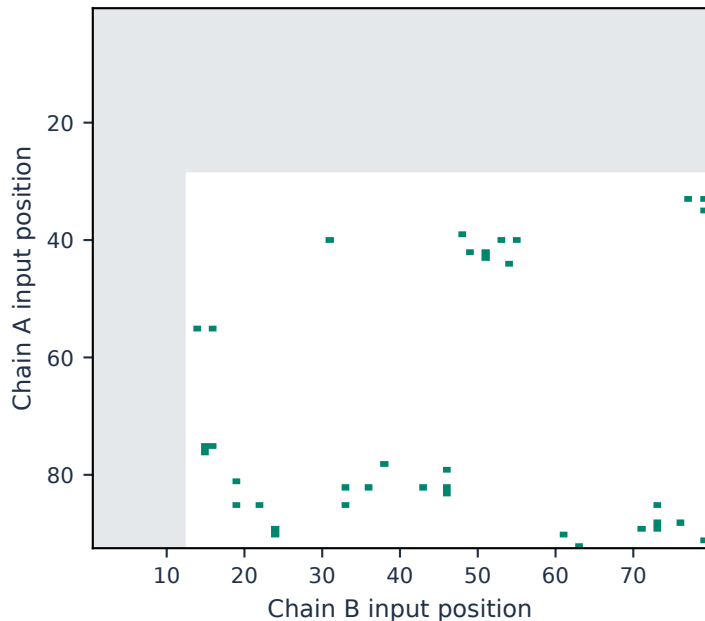
Crystal structure of vaccinia virus G3/L5 sub-complex (SeMet-labeled, P31 space group)

heterodimer | Chains 92 + 79 residues | 37 experimental interface contacts | Test split

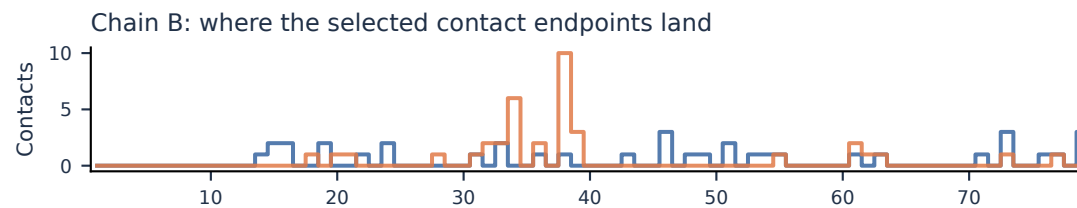
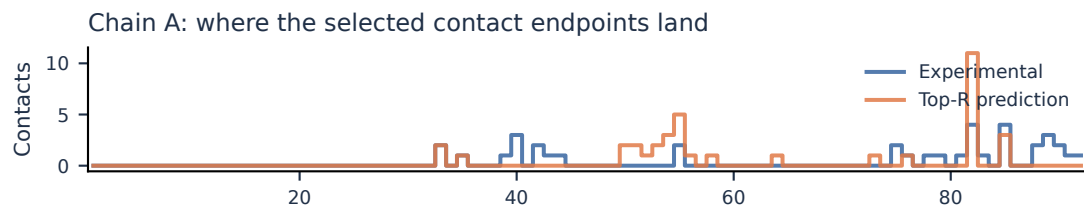
Experimental interface
37 true residue pairs

MarinFold contact frequency
All 1,000 attempted rollouts

Highest-vote 37 pairs (top-R)
0 correct, 37 wrong, 37 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable



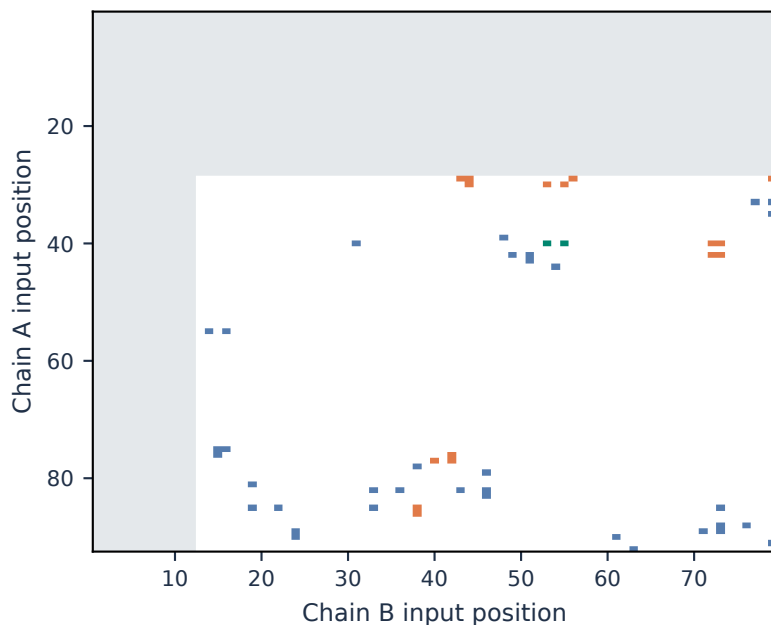
35 of 37 true pairs appear somewhere, but incorrect pairs win the consensus ranking.

Top-R precision = recall = 0.0%. Predicted endpoint residues on the true interface: A 40%, B 53%. True pairs ever sampled: 35/37.

7ytu | What happens inside individual rollouts?

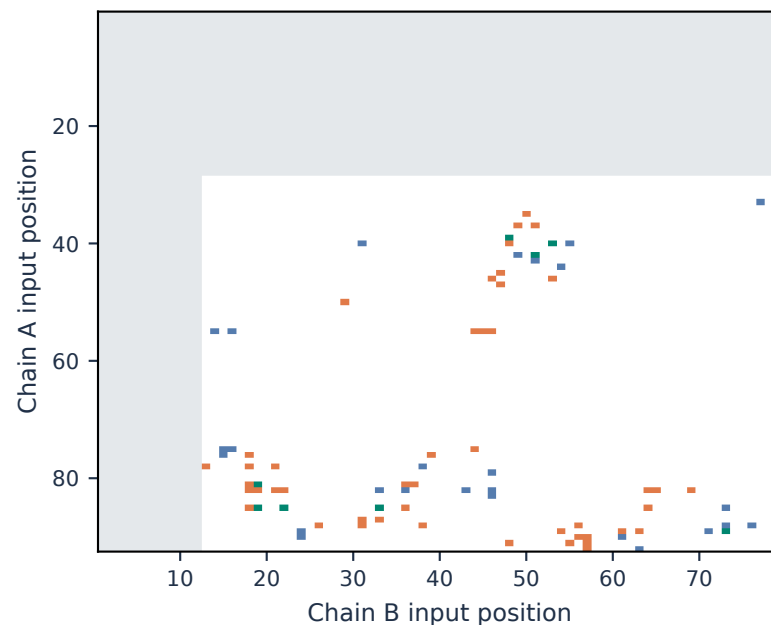
Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.

Contact-count typical | rollout 126
2/18 predicted correct; 2/37 true recovered



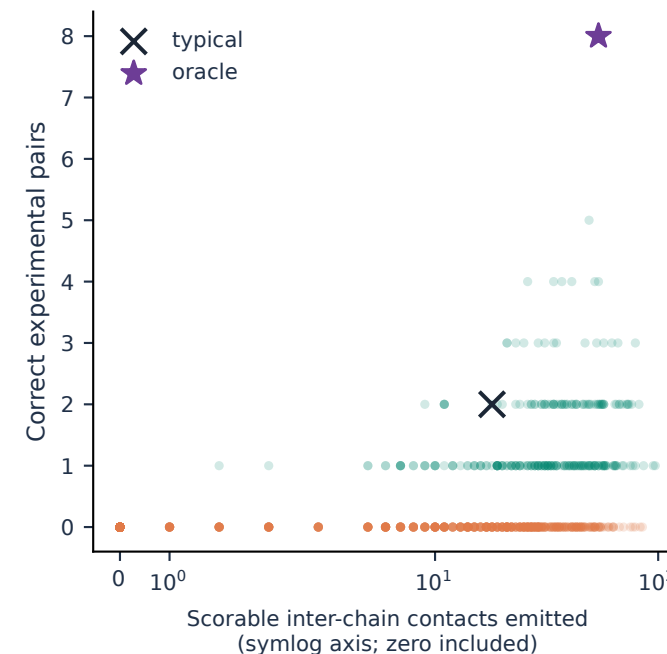
Precision 11.1% | Recall 5.4%

Oracle best@1000 | rollout 978
8/54 predicted correct; 8/37 true recovered



Precision 14.8% | Recall 21.6%

All completed rollouts
Dots overlap; see category counts below



Correct predicted pair

Wrong predicted pair

Missed experimental pair

Unresolved / unscorable

12.2%

No interface
contacts

5.9%

Unscorable
contacts only

55.5%

Scorable, but
all wrong

26.4%

At least one
correct pair

0.0%

Unfinished

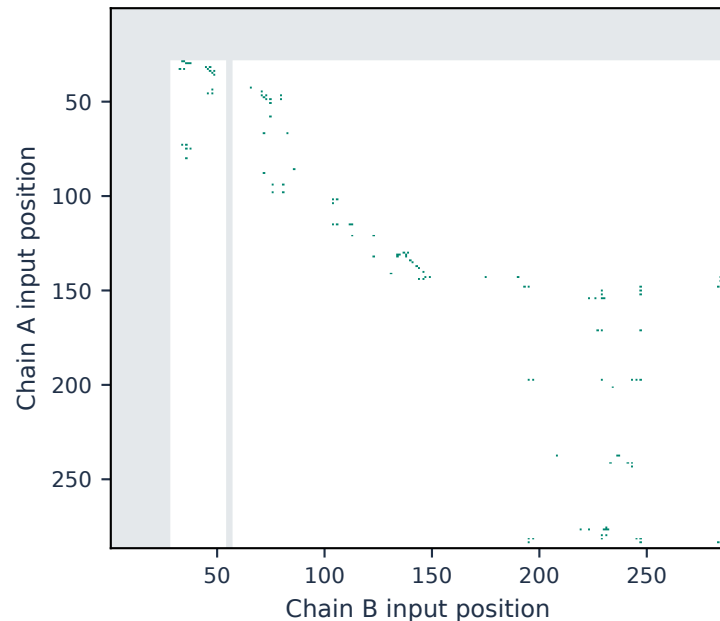
All-rollout median scorable contact count (completed only): 18. Experimental contact count: 37. A green pair is correct; a green-containing rollout is not necessarily a good map.

8onf | Strongest consensus, still many wrong pairs

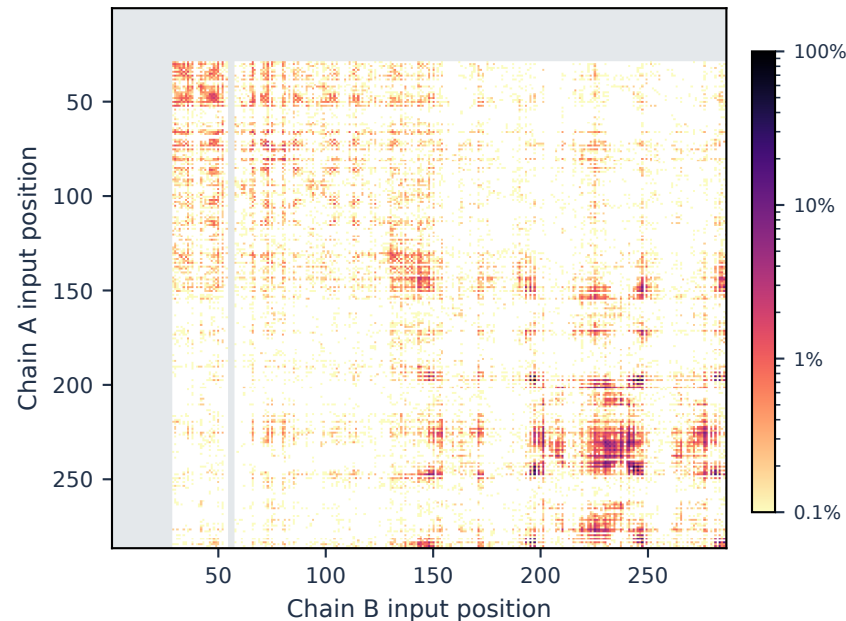
Crystal structure of *Bdellovibrio bacteriovorus* Bd2439 fibre C-terminal domains with ethylene glycol

homodimer | Chains 286 + 286 residues | 122 experimental interface contacts | Test split

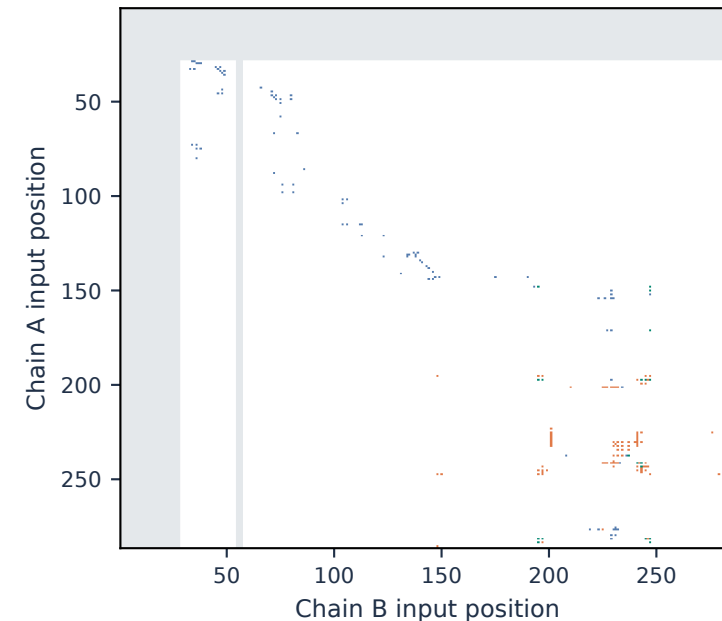
Experimental interface
122 true residue pairs



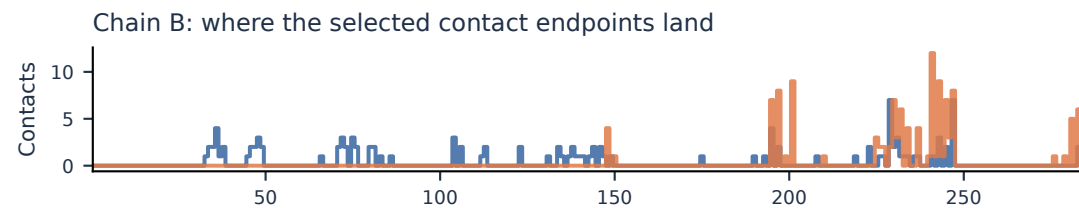
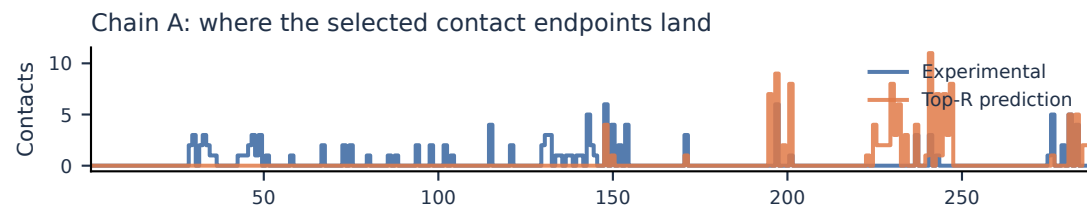
MarinFold contact frequency
All 1,000 attempted rollouts



Highest-vote 122 pairs (top-R)
23 correct, 99 wrong, 99 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable

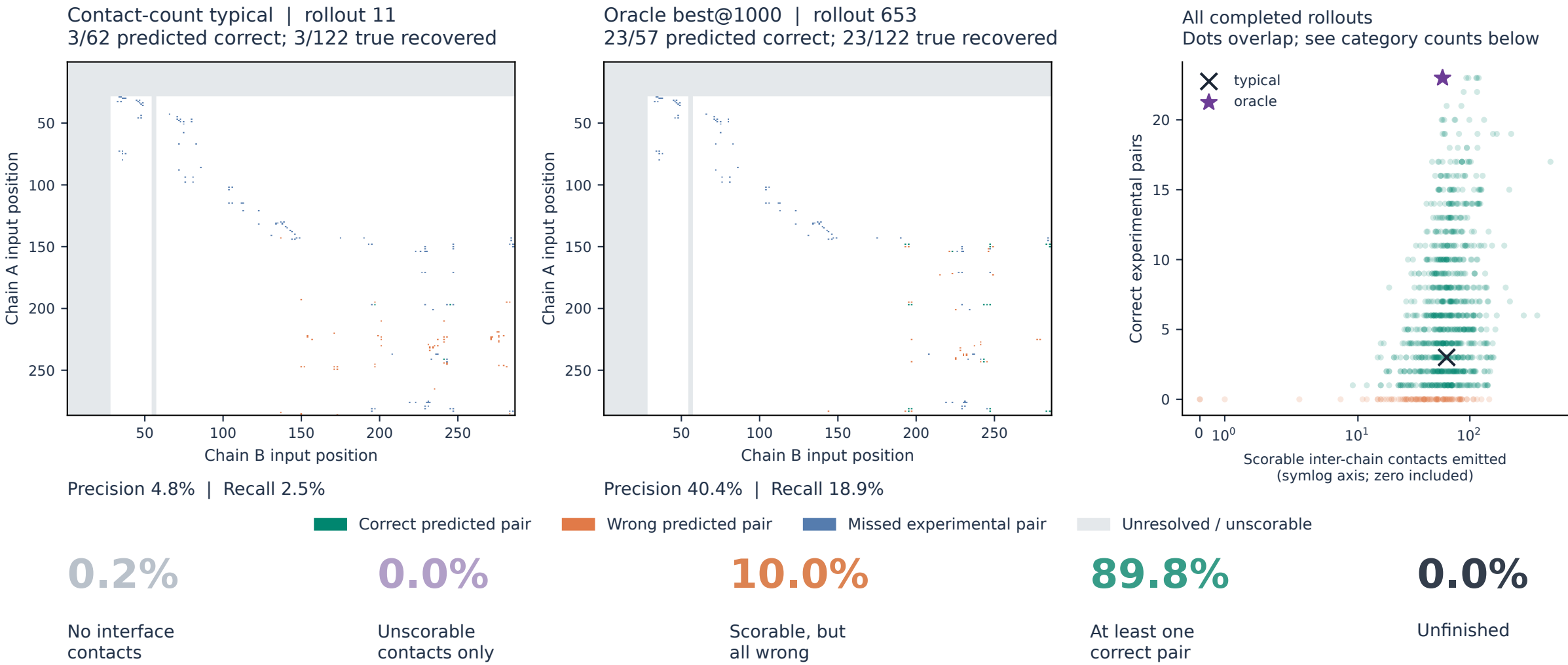


Correct contacts occur often, but votes concentrate on a limited and partly wrong region.

Top-R precision = recall = 18.9%. Predicted endpoint residues on the true interface: A 37%, B 57%. True pairs ever sampled: 118/122.

8onf | What happens inside individual rollouts?

Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.



All-rollout median scorable contact count (completed only): 62. Experimental contact count: 122. A green pair is correct; a green-containing rollout is not necessarily a good map.

8jca | Best individual sample is still incomplete

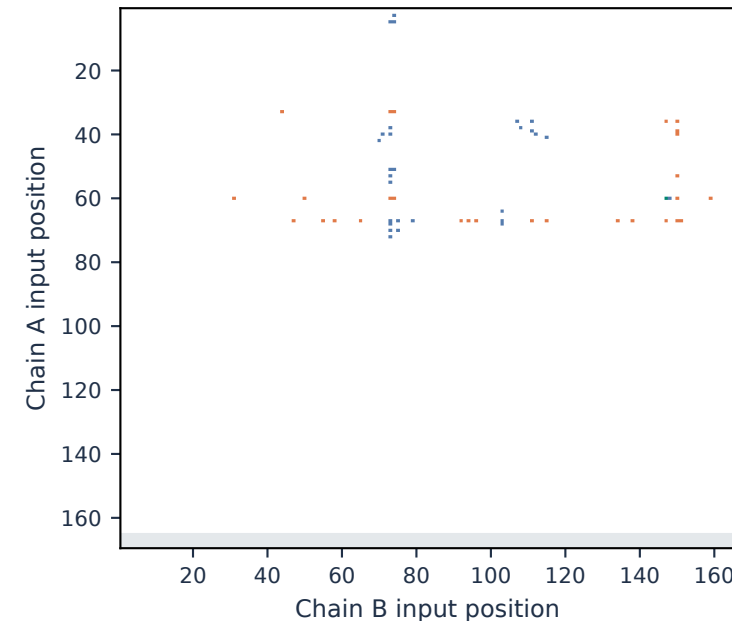
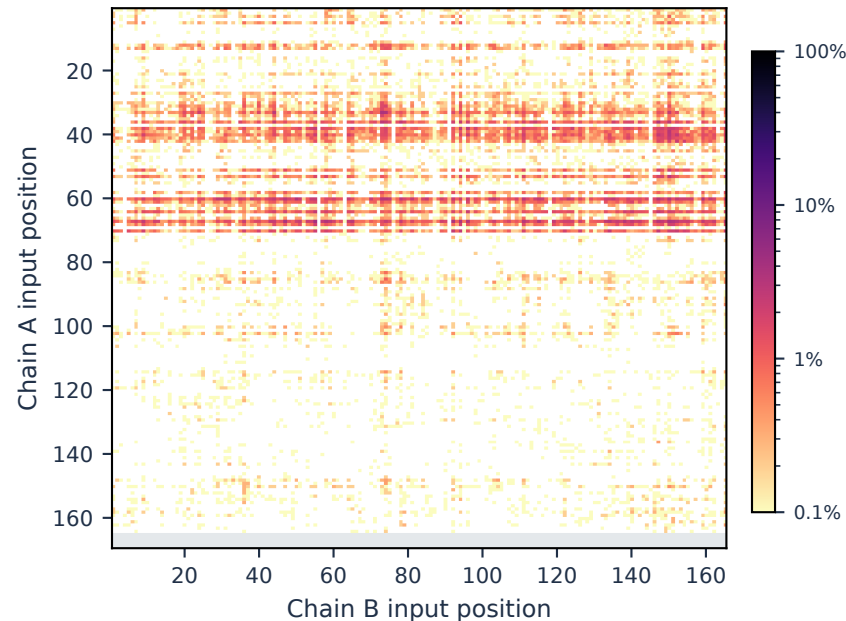
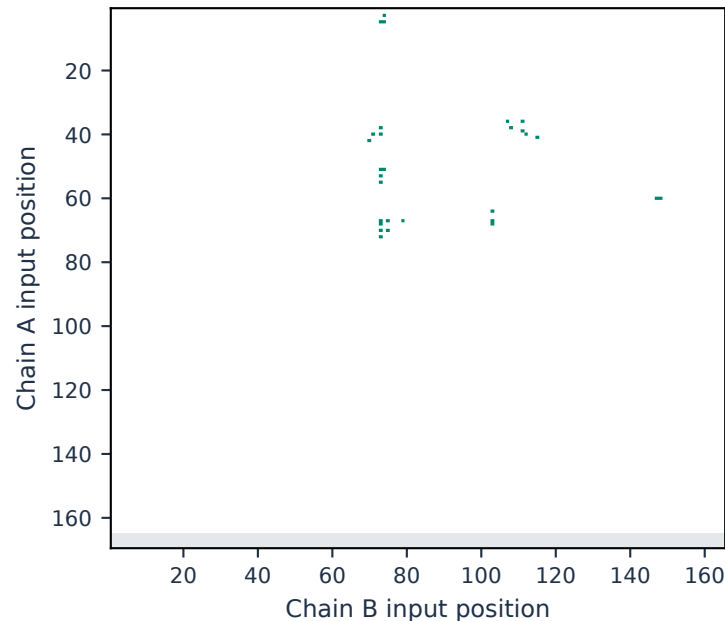
Crystal structure of SKIP RUN domain in complex with GTP-bound Arl8b(Q75L)

heterodimer | Chains 169 + 165 residues | 29 experimental interface contacts | Test split

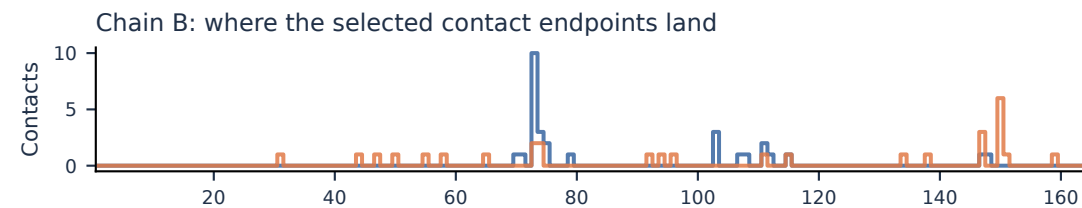
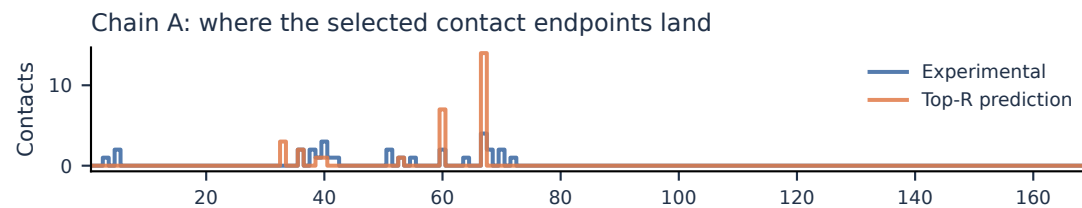
Experimental interface
29 true residue pairs

MarinFold contact frequency
All 1,000 attempted rollouts

Highest-vote 29 pairs (top-R)
1 correct, 28 wrong, 28 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable



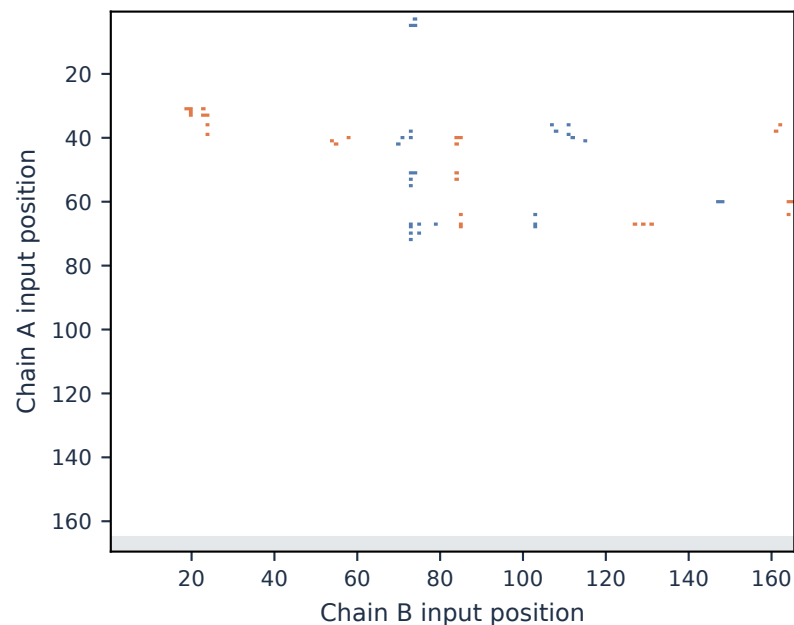
Top-ranked residues often hit chain A's interface, while chain B localization is weaker.

Top-R precision = recall = 3.4%. Predicted endpoint residues on the true interface: A 86%, B 25%. True pairs ever sampled: 28/29.

8jca | What happens inside individual rollouts?

Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.

Contact-count typical | rollout 51
0/28 predicted correct; 0/29 true recovered



Precision 0.0% | Recall 0.0%

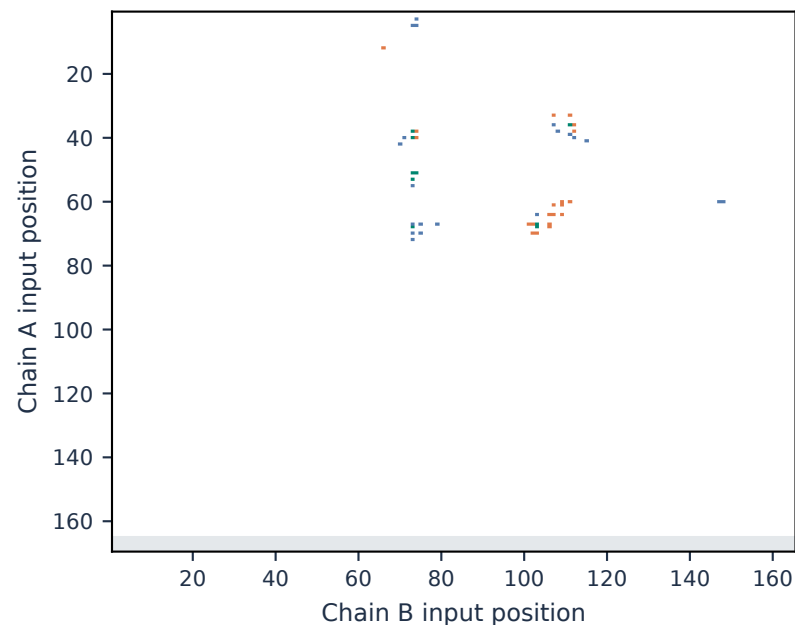
6.8%

No interface
contacts

0.0%

Unscorable
contacts only

Oracle best@1000 | rollout 801
9/29 predicted correct; 9/29 true recovered

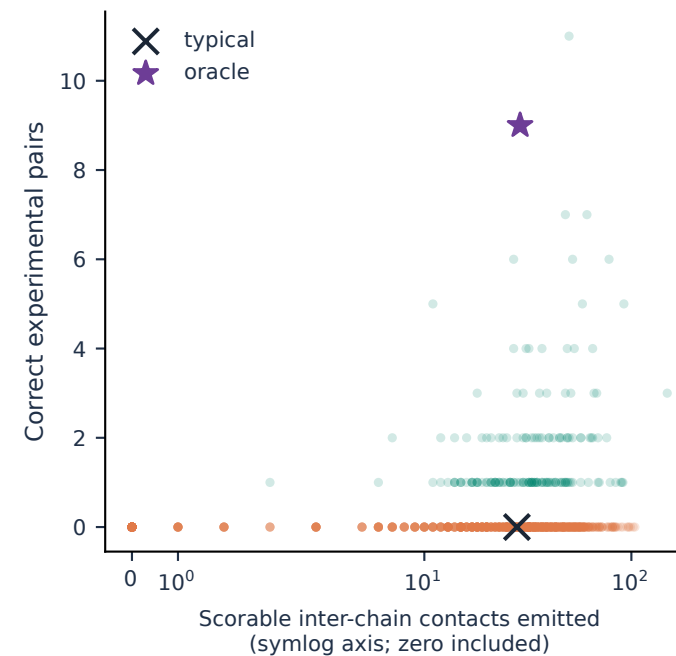


Precision 31.0% | Recall 31.0%

76.8%

Scorable, but
all wrong

All completed rollouts
Dots overlap; see category counts below



16.4%

At least one
correct pair

0.0%

Unfinished

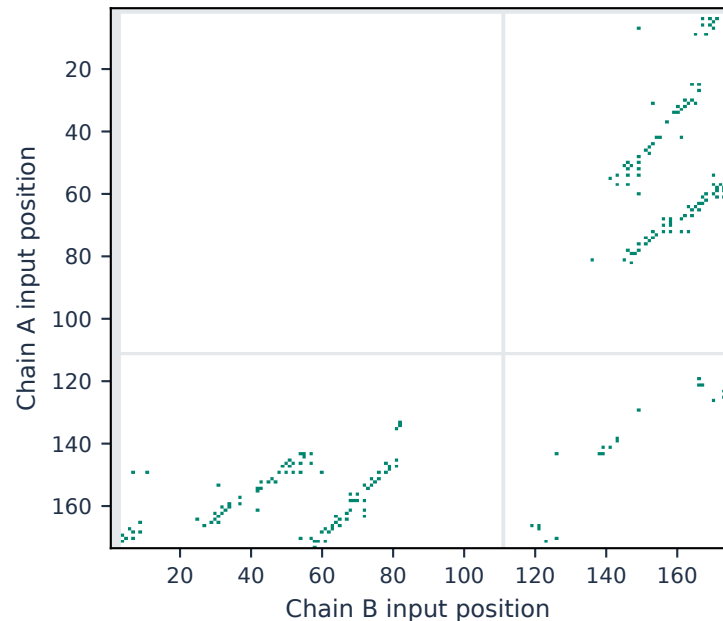
All-rollout median scorable contact count (completed only): 28. Experimental contact count: 29. A green pair is correct; a green-containing rollout is not necessarily a good map.

8smq | Large interface; correct contacts are scattered

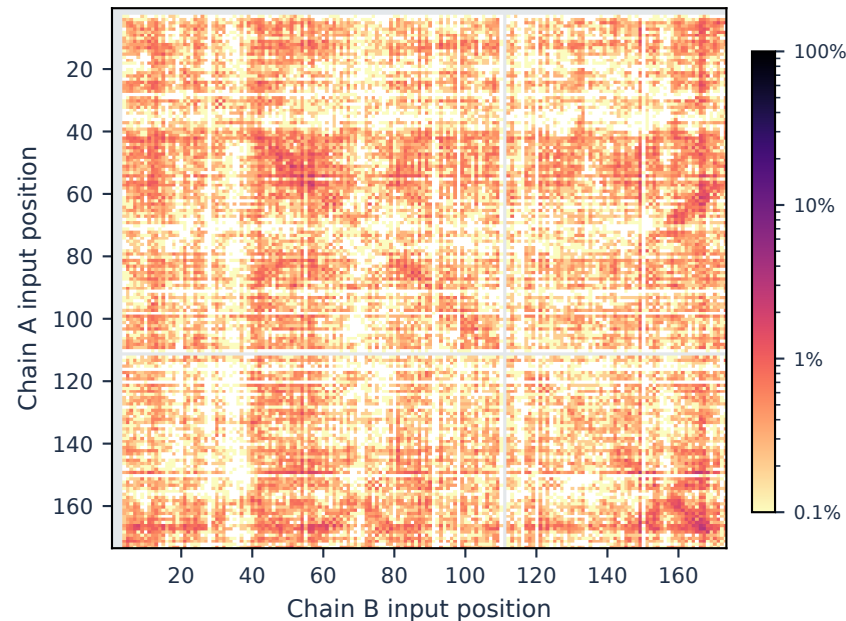
Crystal Structure of the N-terminal Domain of the Cryptic Surface Protein (CD630_25440) from *Clostridium difficile*.

homodimer | Chains 173 + 173 residues | 200 experimental interface contacts | Test split

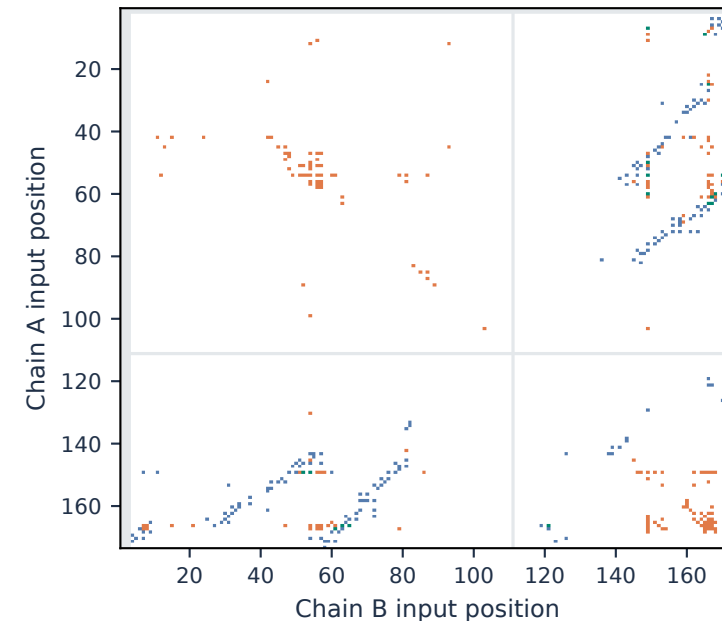
Experimental interface
200 true residue pairs



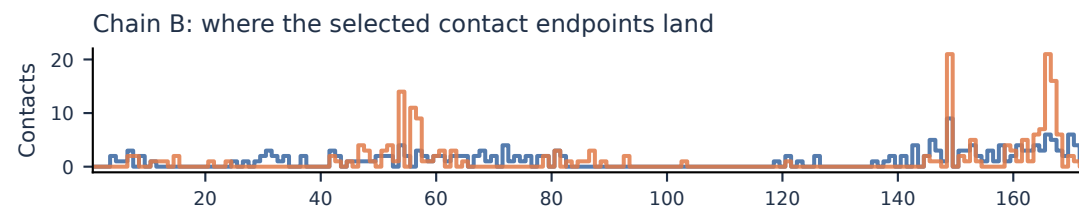
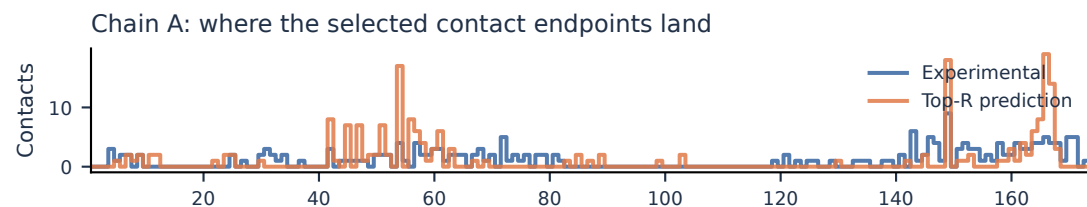
MarinFold contact frequency
All 1,000 attempted rollouts



Highest-vote 200 pairs (top-R)
17 correct, 183 wrong, 183 missed



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable



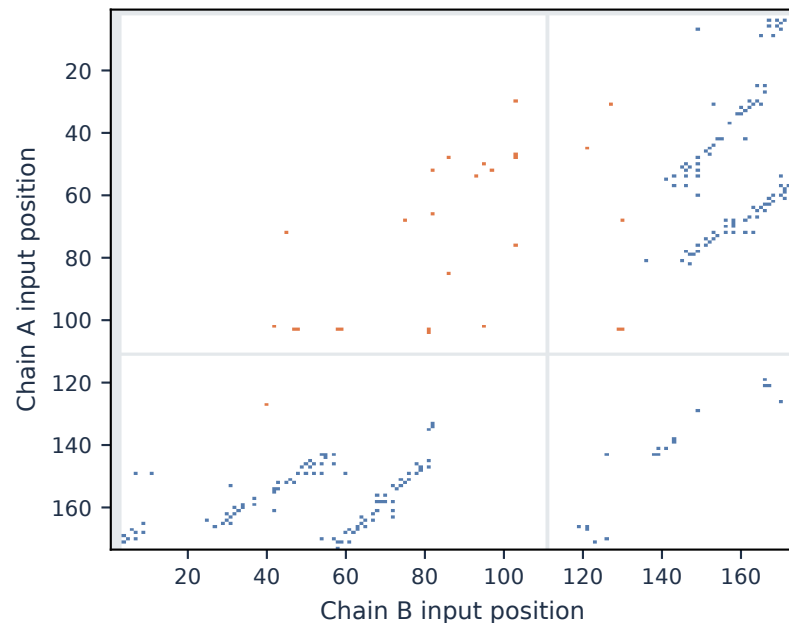
Many predicted endpoints lie on the interface, but most selected residue pairings are wrong.

Top-R precision = recall = 8.5%. Predicted endpoint residues on the true interface: A 70%, B 74%. True pairs ever sampled: 196/200.

8smq | What happens inside individual rollouts?

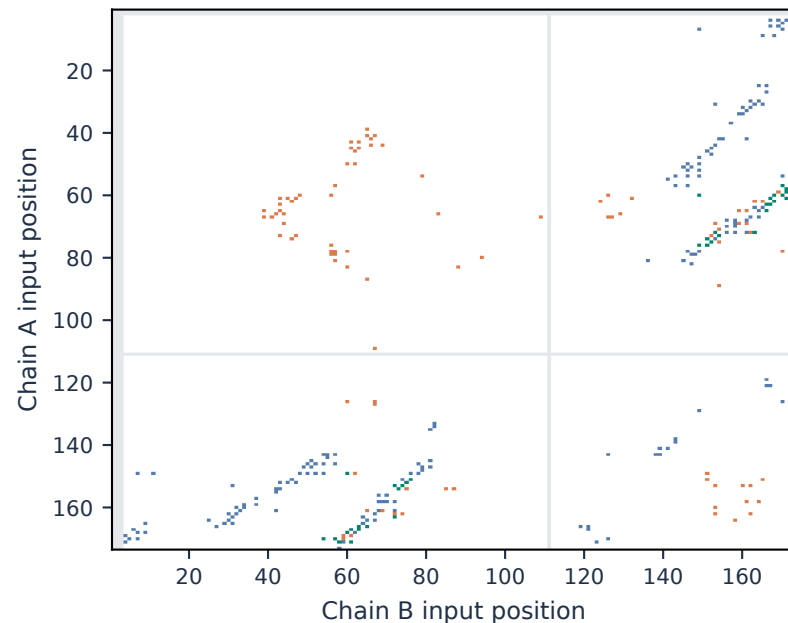
Typical = closest to median scorable contact count, selected without truth. Oracle = highest experimental F1 among 1,000 attempts.

Contact-count typical | rollout 65
0/27 predicted correct; 0/200 true recovered



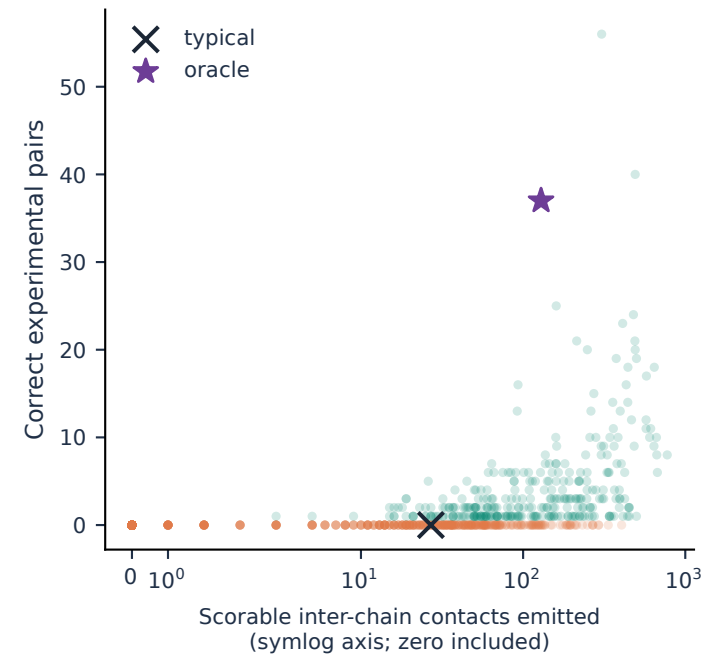
Precision 0.0% | Recall 0.0%

Oracle best@1000 | rollout 347
37/129 predicted correct; 37/200 true recovered



Precision 28.7% | Recall 18.5%

All completed rollouts
Dots overlap; see category counts below



Correct predicted pair Wrong predicted pair Missed experimental pair Unresolved / unscorable

27.3%

No interface contacts

0.1%

Unscorable contacts only

41.5%

Scorable, but all wrong

30.9%

At least one correct pair

0.2%

Unfinished

All-rollout median scorable contact count (completed only): 27. Experimental contact count: 200. A green pair is correct; a green-containing rollout is not necessarily a good map.

Reading the maps and reproducing the atlas

These are diagnostics of contact generation and ranking; no structures were refolded for this report.

Experimental truth

The frozen FoldBench pyconfind interface contacts use degree ≥ 0.001 . Scoring uses only residue pairs with resolved experimental coordinates. This is the same truth and mask as the reported contact evaluation.

Coordinates and colors

Axes are one-based positions within the full input sequence of each chain, not PDB author residue numbers. Unresolved rows/columns are gray. White marks eligible cells with no signal in that panel. Cells are unsmoothed; zoom into the PDF to inspect exact residue pairs. Homodimers use the frozen chain assignment.

Frequency and selection

Frequency is the fraction of all 1,000 attempts that completed and emitted that pair; it is not calibrated confidence. Every frequency panel shares a logarithmic 0.1%-100% scale. Top-R selects exactly as many pairs as there are experimental contacts, with the original stable tie convention. The full-output panels also show unscorable positions.

Individual samples

The typical map is the completed rollout nearest the median scorable contact count; its choice does not inspect correctness. The oracle map maximizes experimental F1, so it is unavailable at inference time. Neither map removes its false positives. Unfinished attempts retain zero credit; their partial maps are excluded.

Interpretation

An absence of inter-chain contacts differs from contacts aimed at unresolved residues, a wrong interface patch, or wrong residue pairing within a partly correct patch. The cohort bars, marginal contact counts and exact-pair error colors separate these cases. The observed six-case patterns are descriptive, not additional blinded benchmark comparisons.

Source: public MarFold bucket, `exp350_foldbench_pair_holdout/contact_eval_v1/sampling_v1`.
Rebuild offline from the committed plotting CSVs: `uv run python build_contact_atlas.py`